

Table 1: Cross-omics Pathway Axis Validation Summary						
Pathway / Axis	Driver Gene	Evidence	N Metabolites	Match Rate	Mean log2FC	Key Metabolites
Polyamine	SAT1	Moderate	7	5/7	−0.52	Ornithine;N8–Acetylspermidine;Putrescine;Spermidine;Spermine;Hypotaurine
Glutamate/cysteine	SLC1A5	Moderate	13	8/13	−0.38	Glutamate;Cysteine;Glutamine;Methionine;Glycine;Serine;Taurine
Antioxidant/NRF2	KEAP1	Moderate	10	5/10	−0.45	GSH;GSSG;NADPH;NADP+;Ascorbate
Inflammation	IL6	Weak	7	3/7	0.28	Prostaglandin E2;Leukotriene B4;Arachidonic acid
Oxidative stress	NOX4	Weak	5	2/5	0.15	Superoxide;H2O2
Lipid signaling	ACSL4	Weak	5	1/5	0.08	Phosphatidylcholine(16:0/20:4)
NAD+ metabolism	NAMPT	Weak	5	1/5	−0.12	NAD+;Nicotinamide
Hypoxia	HIF1A	Weak	3	0/3	0.00	NA